

Task 1: List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

1. Faulty or Failing RAM

Symptom: Continuous long beeps upon booting, frequent Blue Screens of Death (BSODs) with varying error codes, or sudden freezing during memory-intensive tasks.

2. Failing Storage Drive (HDD or SSD)

Symptom: System boots straight to the BIOS or displays a "No Boot Device Found" error; significantly sluggish performance accompanied by clicking or clicking-and-whirring noises (if using an HDD).

3. Overheating CPU / Dried Thermal Paste

Symptom: The computer boots and runs normally for a few minutes, but abruptly shuts down completely (no error message, just sudden loss of power) under heavy loads like gaming or video rendering.

4. Dead or Failing Power Supply Unit (PSU)

Symptom: PC does not start when pressing the power button

5. Defective Dedicated Graphics Card (GPU)

Symptom: Screen displays odd like colored lines, blocks, or flickering shapes during boot or under load, or the monitor receives no signal while the PC's internal fans and lights remain fully active.

Task 2: Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1: Check the wall socket

Plug something else (like a phone charger) into the same socket. If that doesn't work either, the socket is the problem, not the laptop.

Step 2: Look at the charger cable

Check the wire and the plug for any cuts, bends, or burn marks.

Step 3: Check the charger's light

Most chargers have a small light that turns on when it's working. If the light doesn't turn on, the charger itself might be dead.

Step 4: Test the charger with a multimeter

Set the multimeter to DC voltage. Touch the probes to the charger tip (the part that plugs into the laptop). Check if the number matches what's written on the charger (like 19V). If it shows 0 or a very low number, the charger is faulty.

Step 5: Check the charging port on the laptop

Gently wiggle the charger plug while it's inside the laptop's port. If the charging light flickers on and off, the port itself may be loose or damaged inside.

Step 6: Take out and put back the battery

If your laptop lets you remove the battery, take it out, wipe the metal contacts with a dry cloth, then put it back in properly. Sometimes dirty contacts stop the charging.

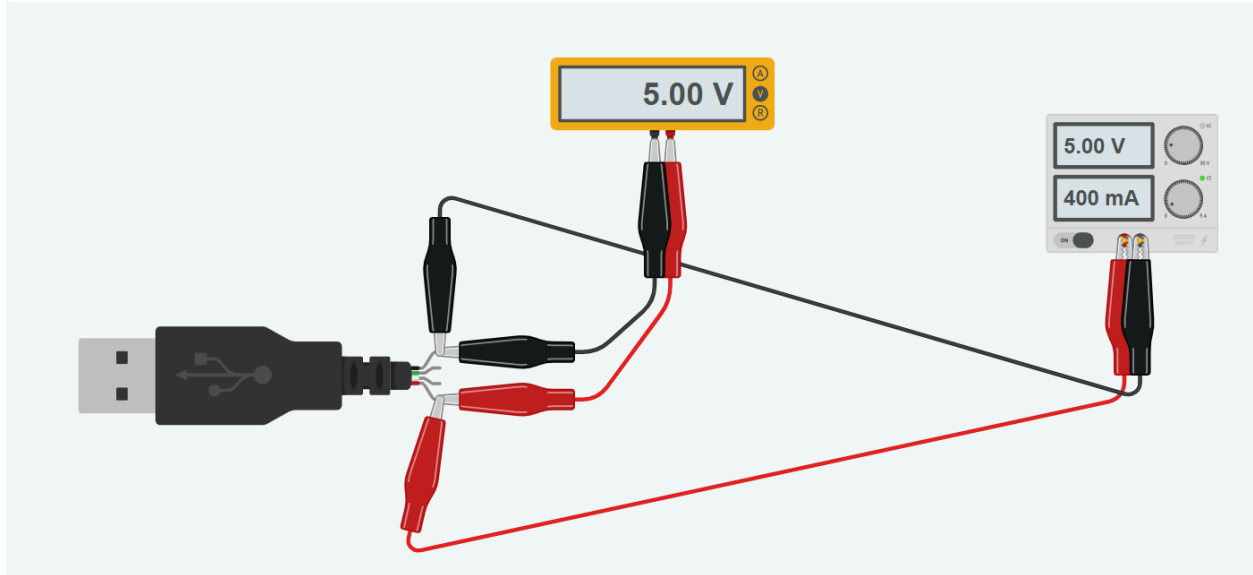
Step 7: Do a hard reset and put back the battery

Unplug the charger, remove the battery if possible, hold the power button for about 20 seconds, then put everything back and try charging again. This clears small glitches.

Step 8: Suspect the battery

If everything else checks out fine but it still won't charge, the battery has likely worn out and needs to be replaced. This is normal after a few years of use.

Task 3: Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).



A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

1. Switch off and unplug the PC from the wall outlet
2. Locate the 24-pin ATX connector running from the PSU to the motherboard. Disconnect it from the motherboard.
3. Set the multimeter to DC voltage, on a range that covers 20V.
4. Jump-start the PSU so it powers on without the motherboard: take a paperclip or short jumper wire and bridge the green wire (PS_ON) to black wire (COM/ground) on the 24-pin connector. This tricks the PSU into thinking the motherboard told it to turn on.
5. Plug the PSU back into the wall and switch it on. The fan should spin up.
6. Probe the yellow and black wires: insert the red probe into the yellow wire's connector slot and the black probe into an adjacent black wire's slot
7. Read the display and record the value.
8. When done, switch off, unplug from the wall, and remove the jumper wire.

Expected voltage

The ATX spec allows $\pm 5\%$ tolerance on the 12V rail, so a healthy reading falls between about **11.4V and 12.6V**.

If the voltage reads below 11.5V

That's outside spec and points toward a failing or degraded power supply, most commonly due to:

- Aging/degraded capacitors in the PSU losing their ability to hold a stable rail under load.
- PSU voltage regulation failure — the voltage drops especially under load (e.g., when the GPU or CPU draw more power)
- Insufficient PSU wattage/capacity for the system's actual power draw, causing the rail to droop when multiple components pull power simultaneously.

The 12V rail is the main power of modern PCs—it delivers power directly to your CPU and Graphics Card (GPU). When these components demand power and the rail drops below the stable threshold, the motherboard's built-in safety mechanisms (Undervoltage Protection, or UVP) instantly trip to prevent data corruption or physical damage, causing the PC to violently shut down without warning.